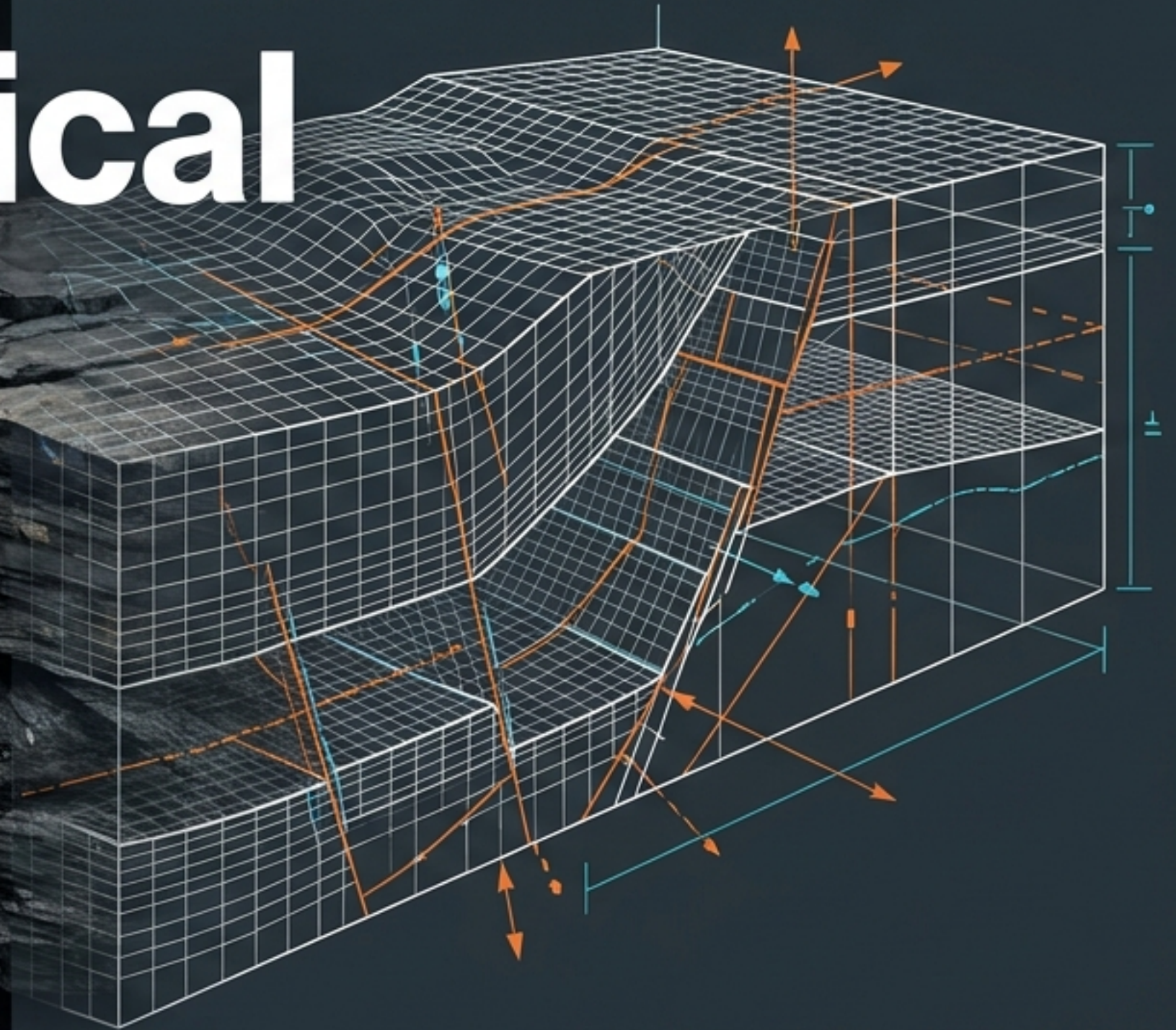


Advanced Geomechanical Modeling

Engineering the Subsurface:
The Architecture of Mechanical
Earth Models (MEM)

Prepared by Engineer: Sajjad K.Hano



The Definitive Digital Twin

A **Mechanical Earth Model (MEM)** translates physical subsurface chaos into predictable numerical behavior. It empowers engineers to simulate state of stress, pore pressure, and rock deformation before a drill bit ever touches the ground.

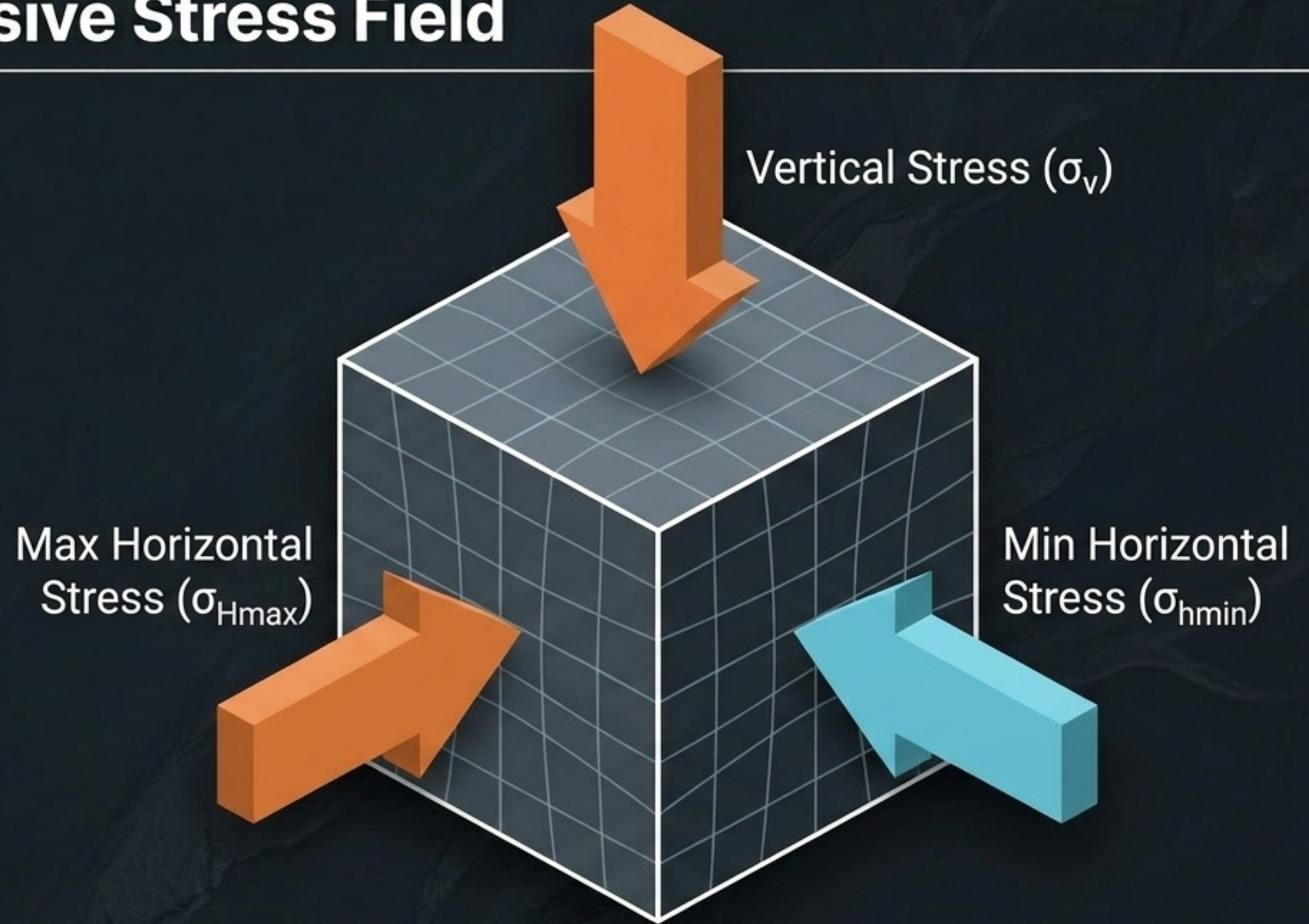


The 3D Compressive Stress Field

The subsurface is governed by three mutually orthogonal principal stresses.

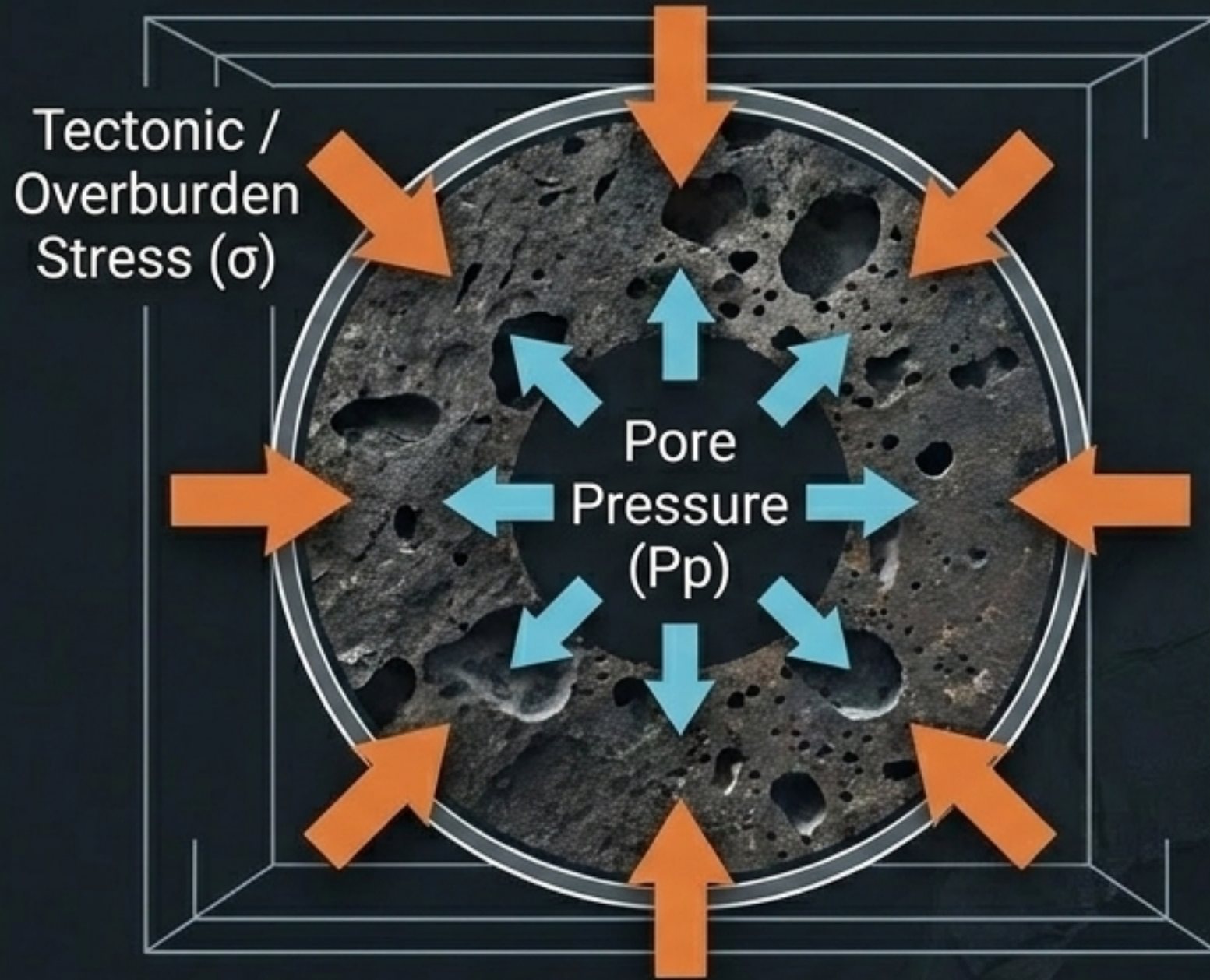
The contrast between these forces dictates everything from hydraulic fracture orientation to structural shearing.

The minimum horizontal stress dictates the precise pressure required to fracture a formation.



The Battle Within the Rock

$$\sigma' = \sigma - \alpha P_p$$



Rocks only experience **Effective Stress (σ')**. Pore pressure acts as an internal shield against confining tectonic forces. When pore pressure drops, effective stress spikes—driving permanent deformation and structural failure.

Decoding Rock Behavior Under Stress

ELASTICITY (Deformation)

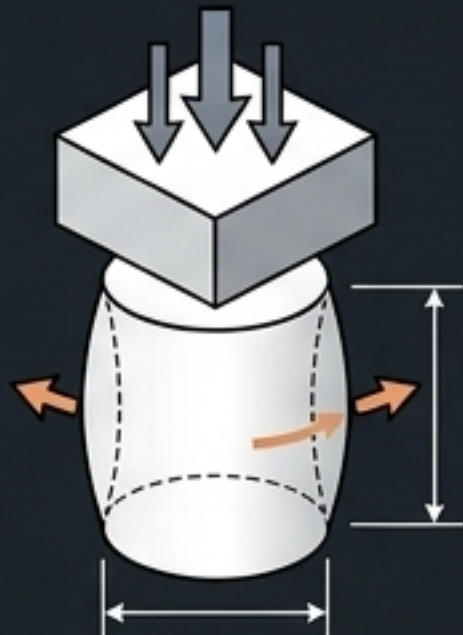
Young's Modulus (E)

Rock stiffness (ratio of axial stress to strain).



Poisson's Ratio (ν)

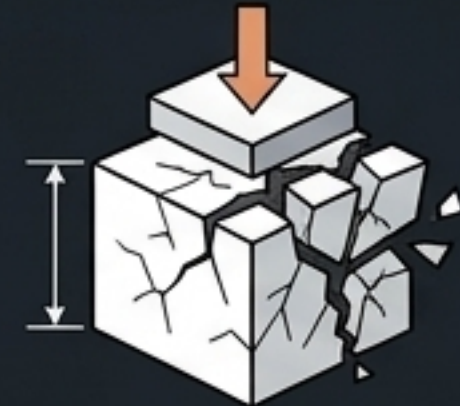
Lateral expansion under compression.



STRENGTH (Destruction)

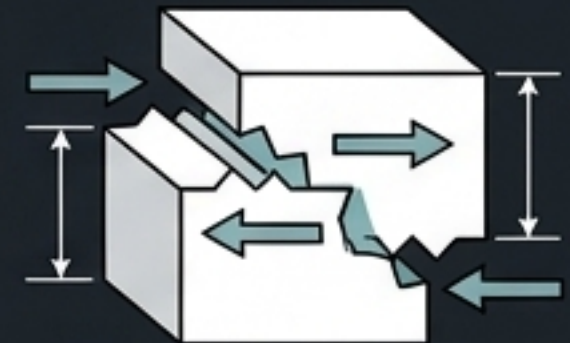
Unconfined Compressive Strength (UCS)

Max compressive bearing without confinement.



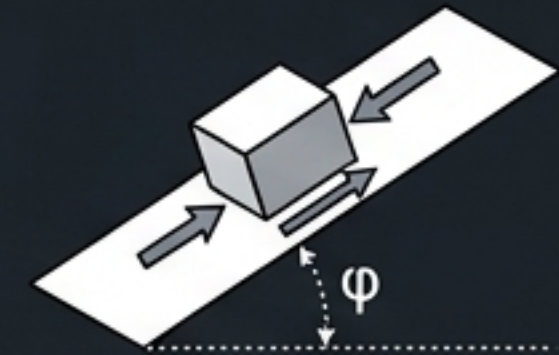
Cohesion (C)

Inherent shear strength at zero normal stress.



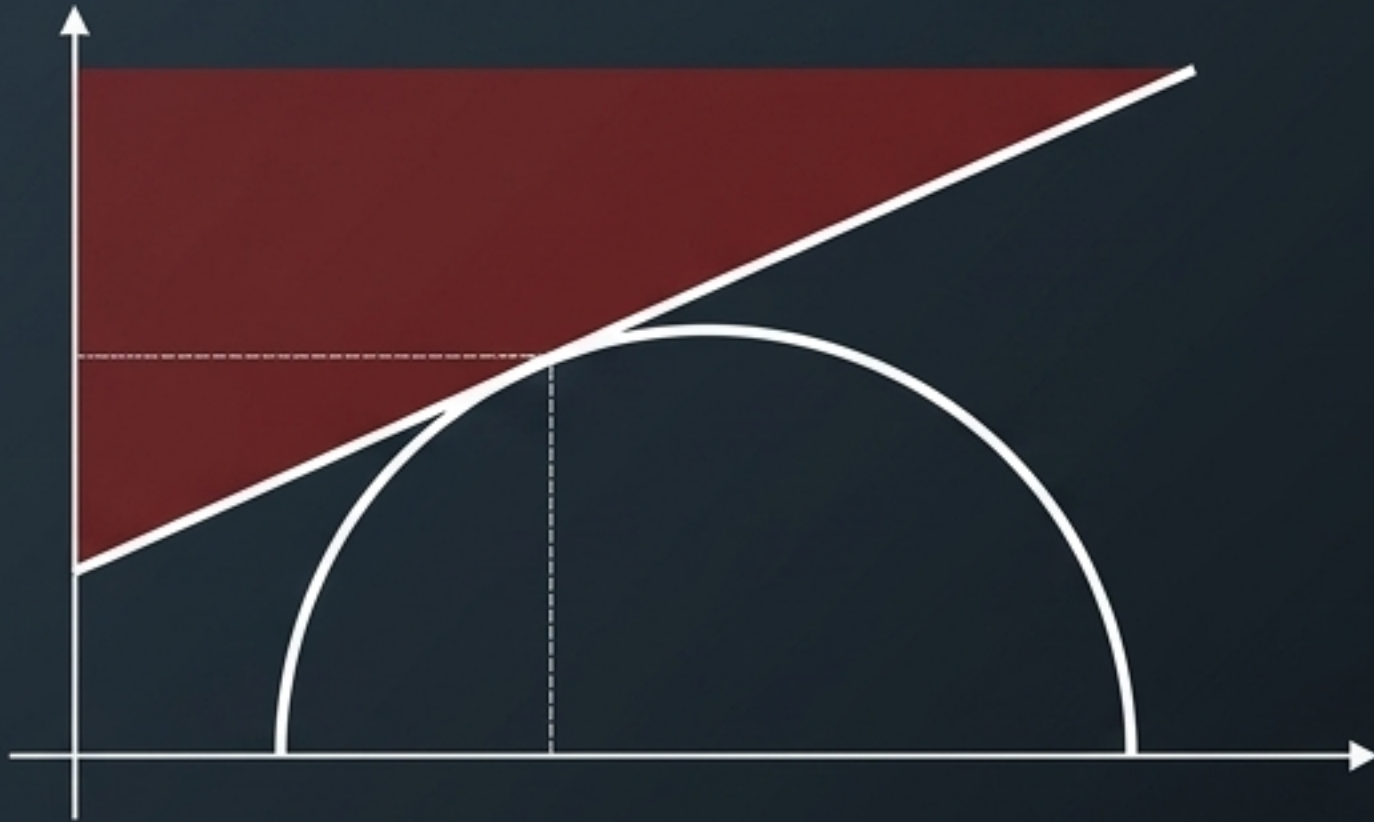
Internal Friction Angle (ϕ)

Resistance to shearing along a failure plane.



Predicting Failure Before It Happens

Shear Breakouts:



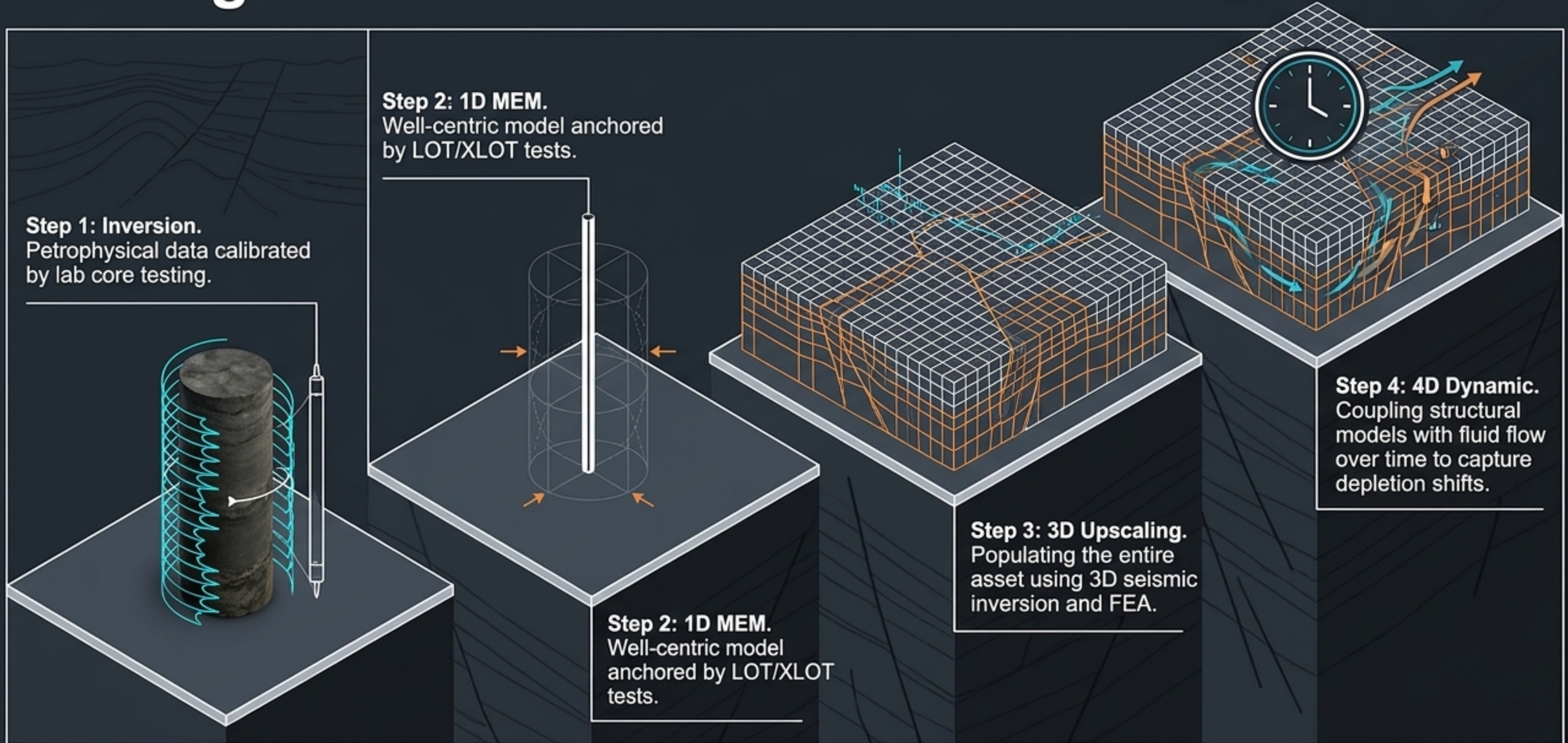
Governed by the **Mohr-Coulomb Criterion**. When operating shear stress exceeds intrinsic shear strength, the rock shears, causing catastrophic wellbore collapse.

Tensile Fracturing:



When minimum effective principal stress drops below the rock's tensile strength, the rock rips apart, leading to unintentional fluid loss.

Scaling the Model: From Borehole to Basin

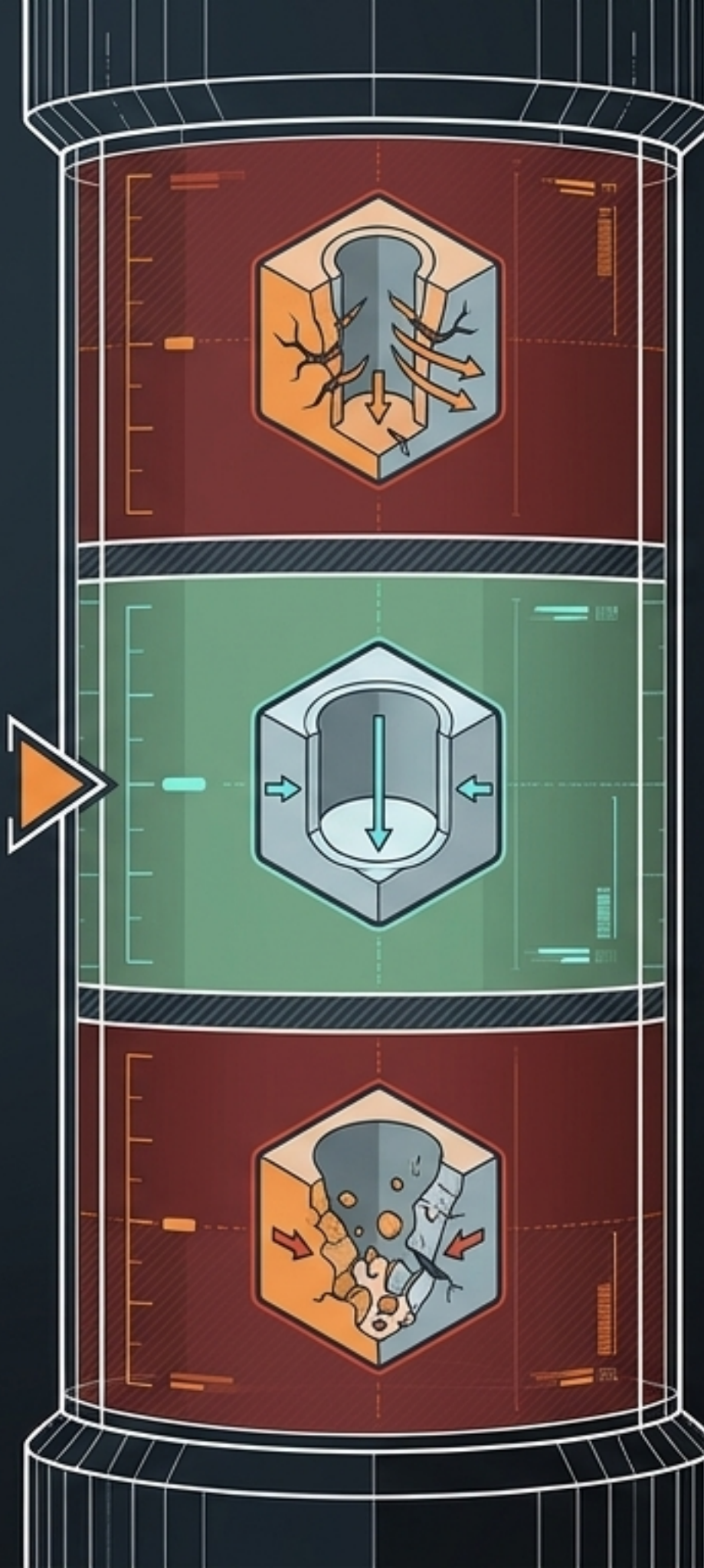


Application I: The Safe Mud Weight Window

Drilling removes load-bearing rock, creating concentrated hoop stresses.

A highly calibrated MEM defines the exact fluid weight required to balance the open hole—preventing both catastrophic structural collapse and expensive drilling fluid losses.

Optimal MW



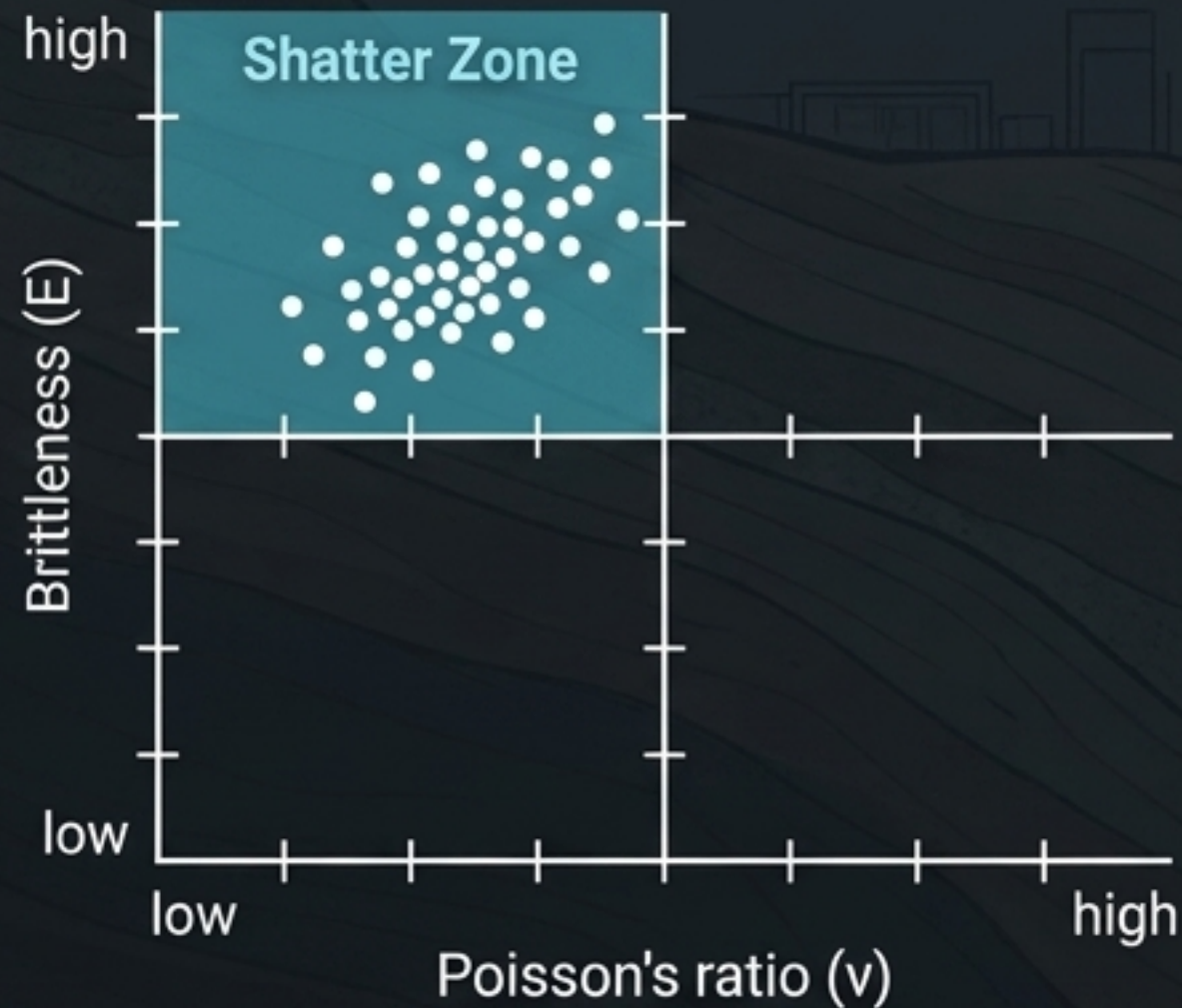
Fracture Gradient
(Risk of Fluid Losses)

The Safe Mud Weight Window

Pore Pressure / Shear Collapse Gradient
(Risk of Breakouts)

Application II: Managing Extraction and Compaction

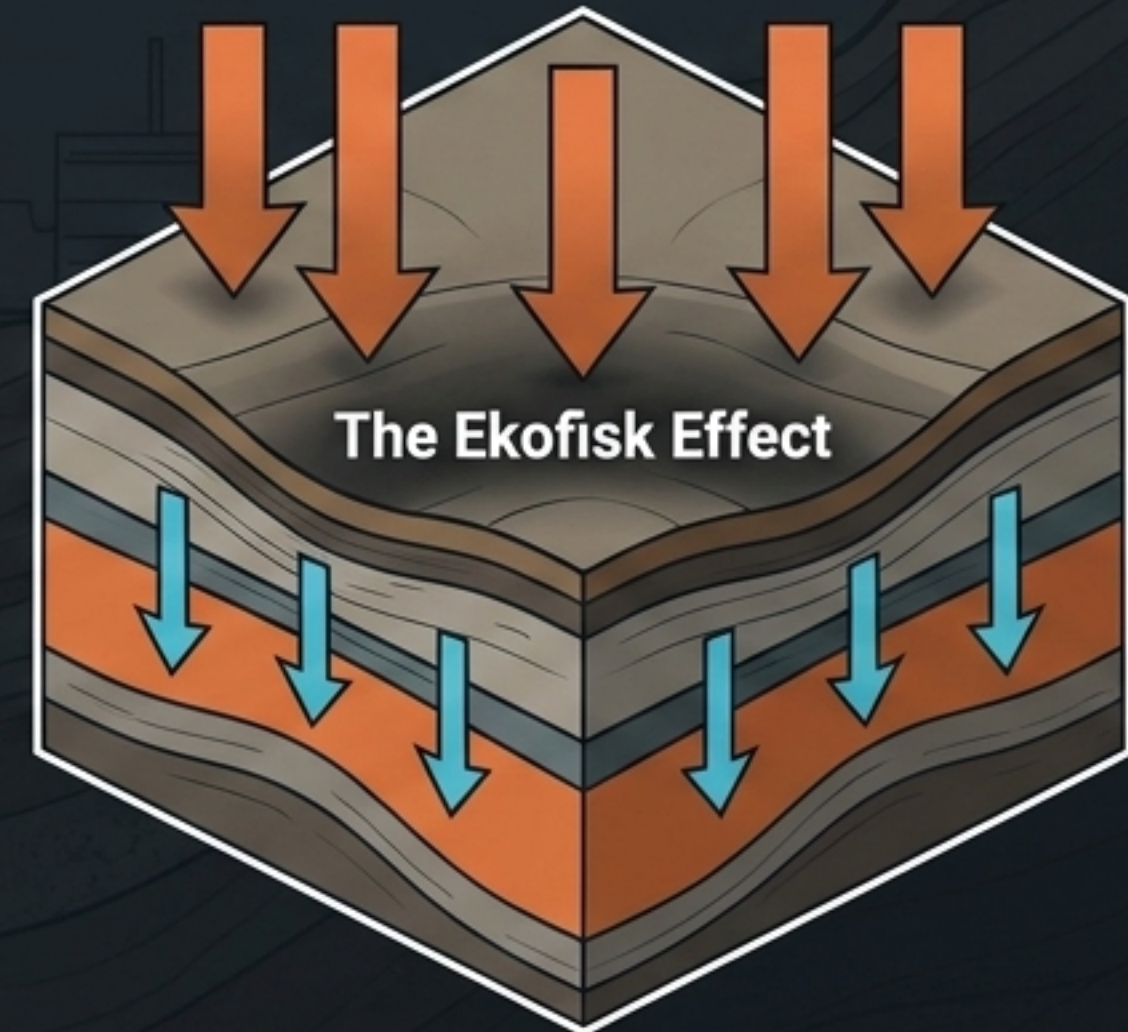
Fracability - Breaking on Purpose



Unconventional Fracking:

Identifying stress barriers and fracable zones that shatter cleanly rather than deform plastically.

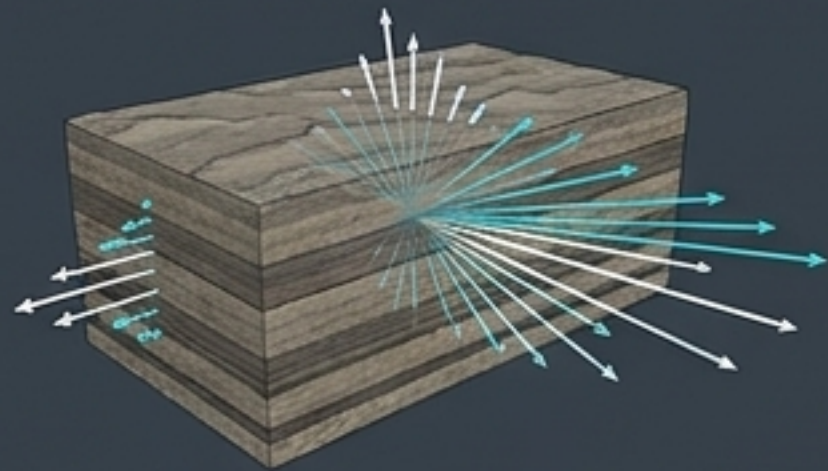
Subsidence - Breaking by Accident



Subsidence & Faulting:

Tracking how reservoir depletion drops pore pressure, spiking effective vertical stress. This triggers severe compaction and reactivates sealing faults.

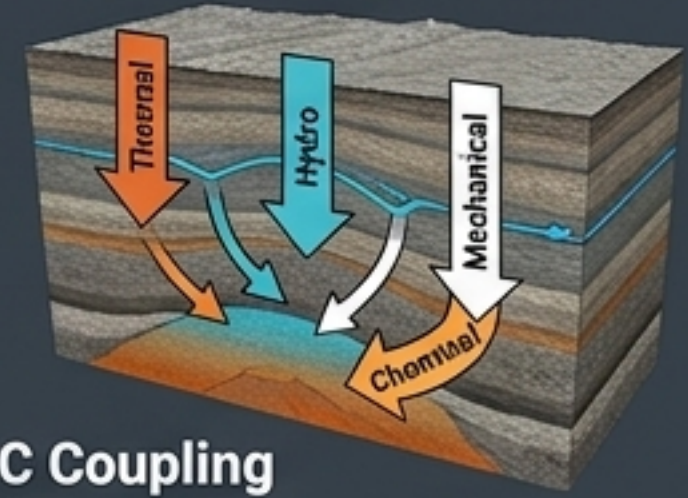
The Next Evolution of Geomechanics



Shale Anisotropy

Modeling directional property variations in complex layered rocks.

Advanced MEM



THMC Coupling

Thermal, Hydro, Mechanical, Chemical integration for Deep Geothermal and CO2 Storage.



Machine Learning

AI algorithms mapping real-time drilling data directly back into the MEM.

Where is your asset's vulnerability? Optimizing mud weights for highly deviated wells, or modeling fault reactivation for Carbon Storage?